Soil Analysis & Crop Recommendation System

(AgroSmart — AI-Based Soil Analysis & Crop Recommendation)

Minor Project Part-I | Panel Evaluation 1

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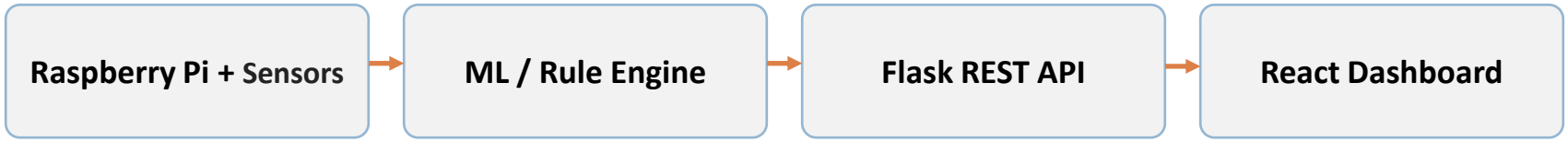
Introduction

Overview

- AgroSmart is a full-stack web application connected to a Raspberry Pi 5 soil-analysis unit.
- It reads real-time soil sensor data (moisture, pH, temperature, N-P-K), classifies soil type using a CNN, and produces recommendations for crop selection, fertilizer and irrigation.
- Tech stack: Python Flask, SQLite, scikit-learn, TensorFlow/Keras, React, Vite, Tailwind CSS.

Objective

- To build an affordable, sensor-driven decision-support tool that helps a farmer choose a suitable crop, fertilizer and irrigation schedule from real soil data instead of guesswork.



Problem Statement

- Most small and mid-scale farmers assess soil health visually or by experience, without measuring moisture, pH or nutrient (N-P-K) levels.
- **This leads to:**
  - Wrong crop choice for the actual soil condition.
  - Over- or under-irrigation, wasting water or stressing the crop.
  - Incorrect or excess fertilizer use, raising cost and harming soil over time.
- Low-cost IoT sensors and machine learning now make it possible to measure soil parameters continuously and turn them into a specific, actionable recommendation.

Why it matters: crop and water losses from poor soil decisions directly affect farmer income and are largely preventable with real-time, data-driven guidance.

Literature Review

Study	Approach	Limitation / Gap
Doshi et al., AgroConsultant, IEEE ICCUBEA, 2018 [1]	ML-based crop recommendation using soil, weather and rainfall data	Uses historical/dataset-based soil and environmental parameters rather than live IoT sensor readings.
Chatterjee et al., Soil Image CNN, IEEE CCCI, 2021 [2]	Compares CNN architectures (ResNet50, MobileNetV2, VGG) for soil-image classification	Classifies soil type only; does not extend to crop/fertilizer/irrigation advice
Gyasi & Purushotham, Soil-MobiNet, Sensors, 2023 [3]	Lightweight MobileNet-based CNN for on-device soil classification	Focuses on image-based soil classification and geospatial identification; it does not combine soil-image classification with live soil sensor data and crop/fertilizer/irrigation recommendations.

Research Gap: Existing studies largely address individual components such as crop recommendation or image-based soil classification. AgroSmart aims to integrate live soil sensing, CNN-based soil classification, crop/fertilizer recommendation and rule-based irrigation guidance into a single low-cost decision-support platform.

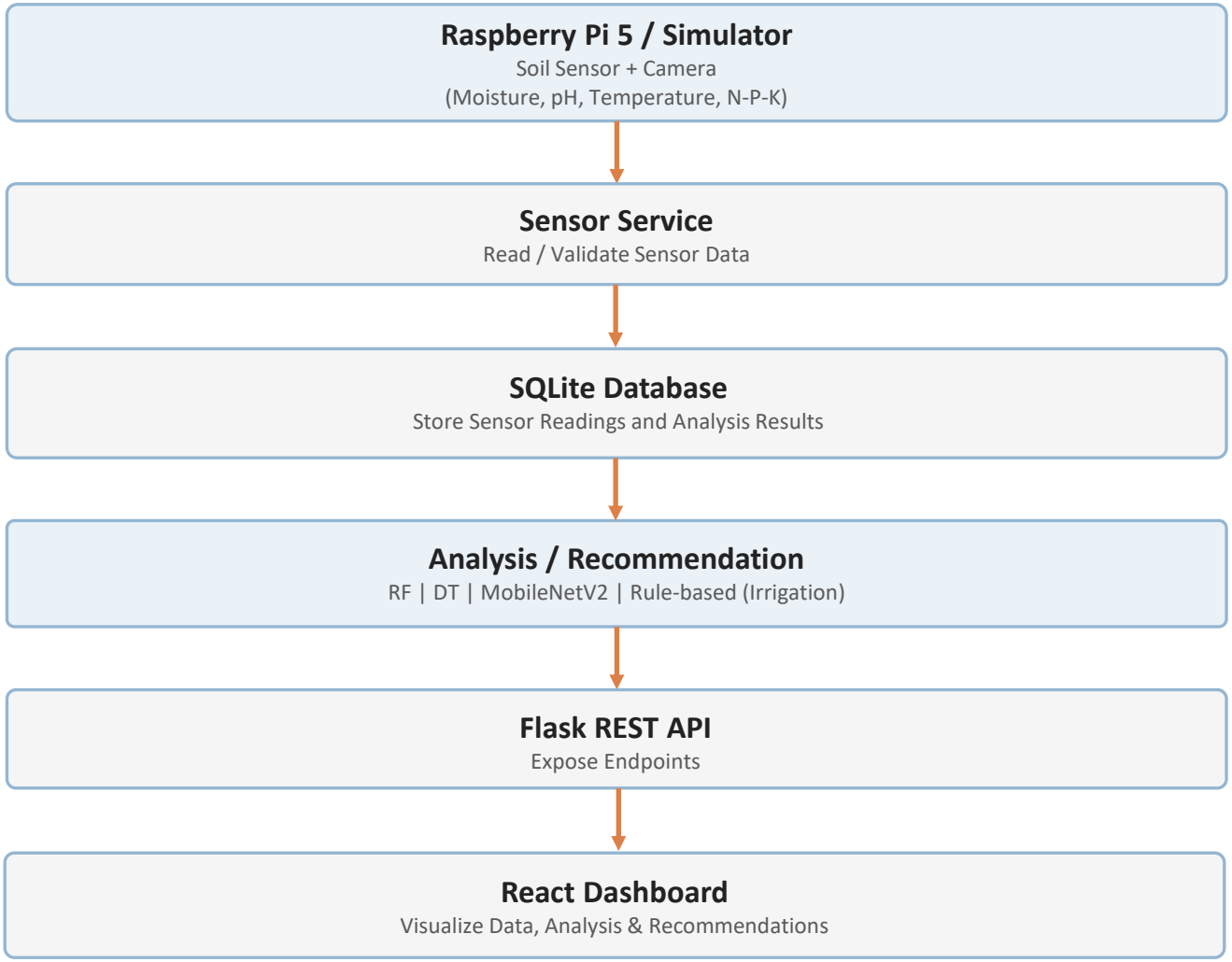
Work Plan (Weekly Progress)

Week	Duration	Key Activity	Status
Week 1	21–25 Jul	Studied objectives; collected & preprocessed soil dataset; selected ML algorithm; built & tested initial crop model	**Completed**
Week 2	27 Jul–01 Aug	Studied IEEE papers on soil analysis & smart agriculture; reviewed Raspberry Pi / camera / DHT sensor docs; finalized system design	**Completed**
Week 3	03–08 Aug	Studied smart-agriculture applications (irrigation, pest/disease detection, farm management) to scope features	**Completed**
Week 4	10–15 Aug	Backend, frontend and README development started; Raspberry Pi 5 setup & connectivity	**Completed**
Week 5	17–22 Aug	Continued documentation; reviewed research and planned features; verified Pi 5 connectivity	**Completed**
Week 6	01–05 Sep	REST API implementation; trained model on Kaggle demo data; API testing	**Completed**
Week 7 onward	Ongoing	Real sensor-hardware integration, full test-suite run, model evaluation & report	**In Progress**

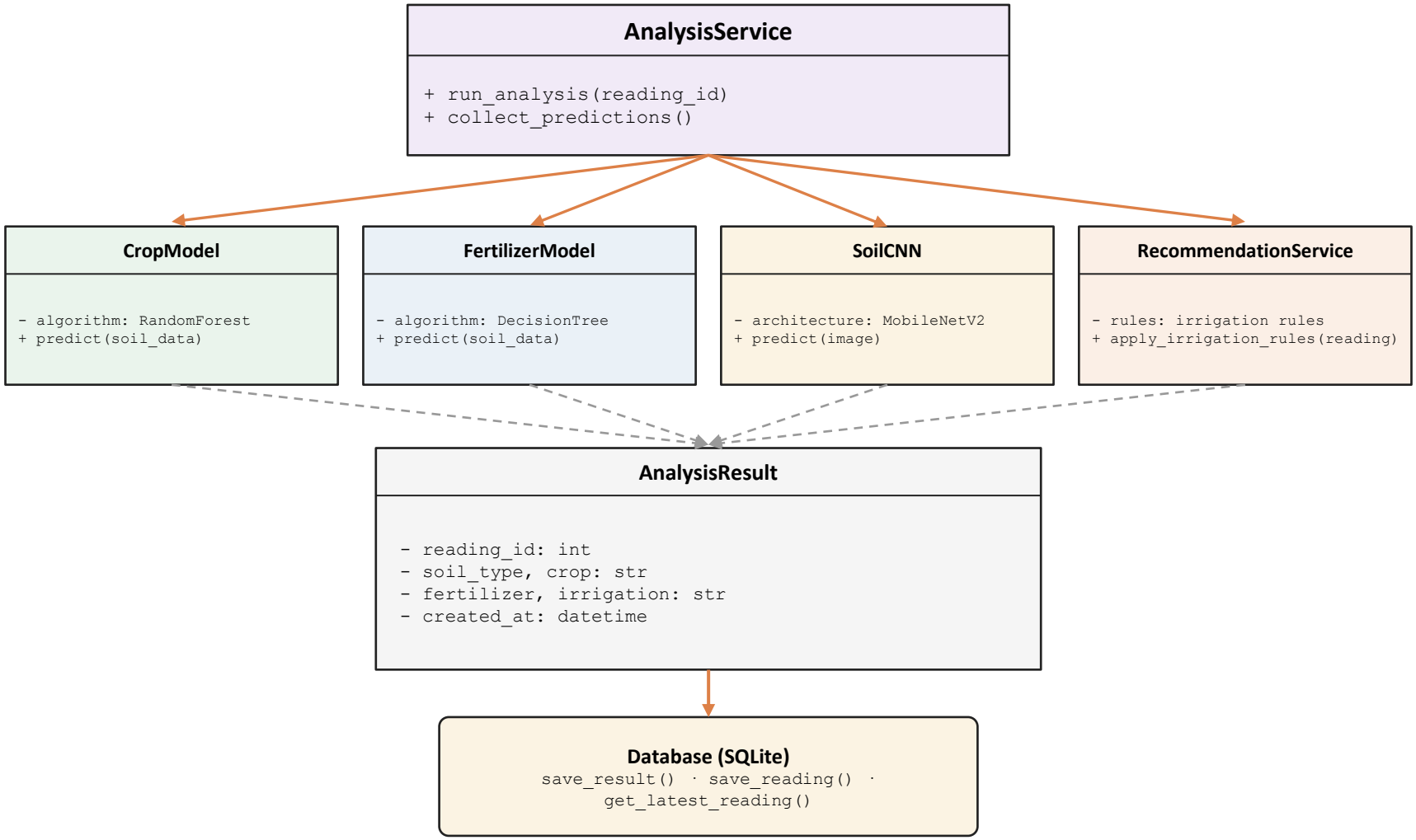
Individual Contribution

Member	Enrollment No.	Contribution
Dharmendra Meena	241B616	Backend development using Flask, ML model development, API integration, React frontend development
Mayur Kushwah	241B623	Deployment and environment configuration, API/backend deployment, integration testing, documentation and project presentation
Shivani Yadav	241B315	Project coordination, literature survey, requirement analysis, system planning and documentation

Work Done So Far — System Architecture

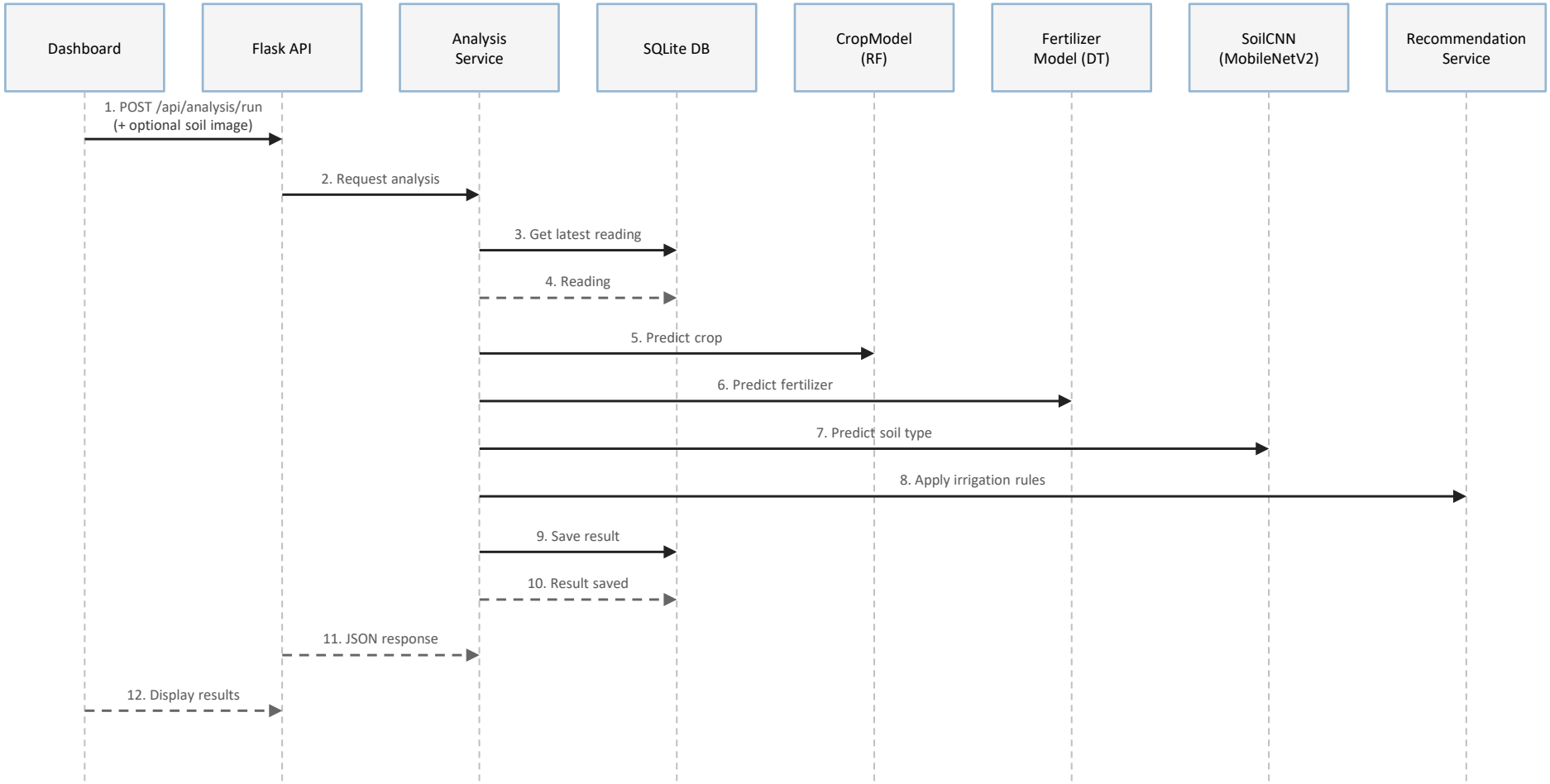


Work Done So Far — UML Class Diagrams

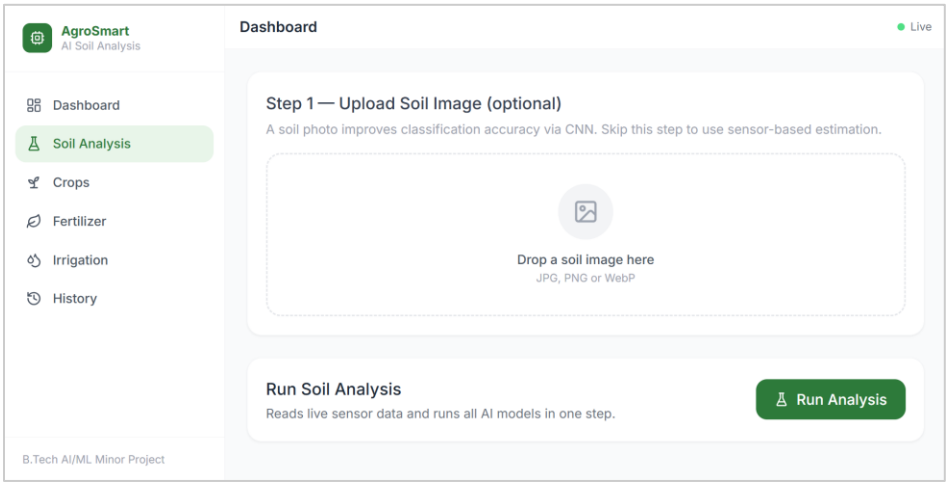


Work Done So Far — Sequence Diagram

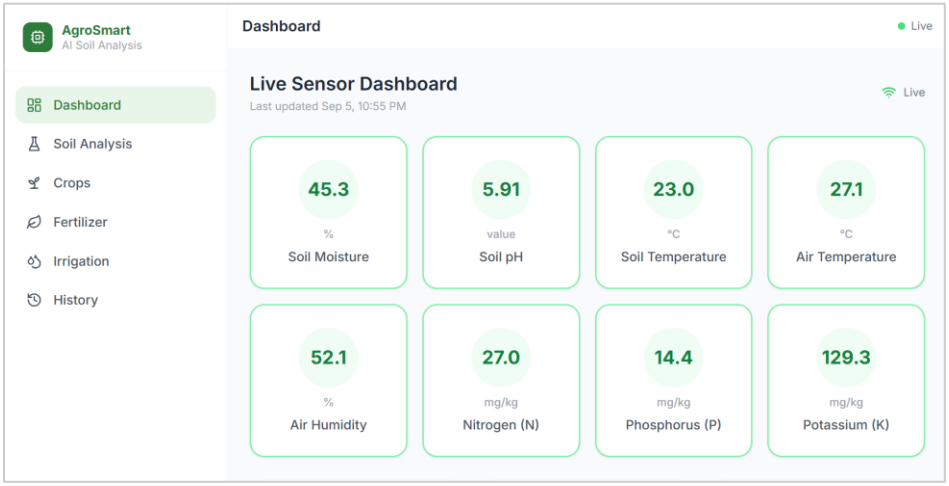
"Run Soil Analysis" — POST /api/analysis/run



Work Done So Far — Application UI

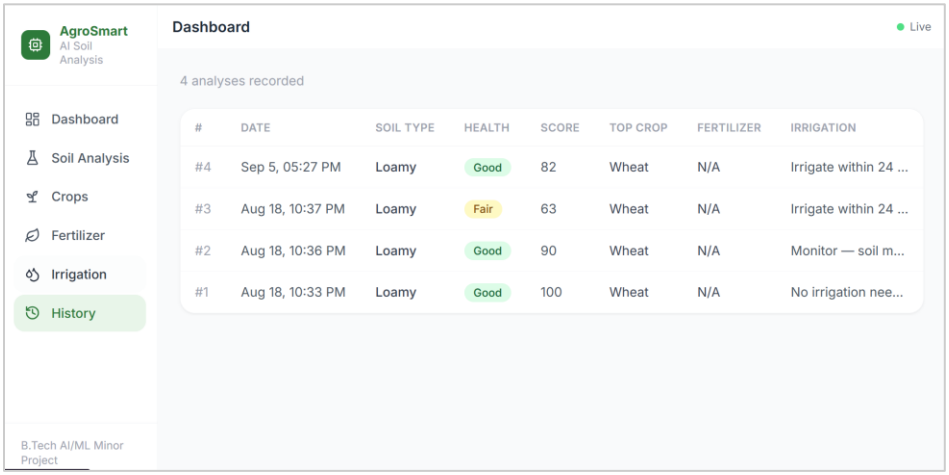


- **Soil Analysis page**
- Lets the user optionally upload a soil photo (for CNN classification) and run the full AI pipeline in one click.

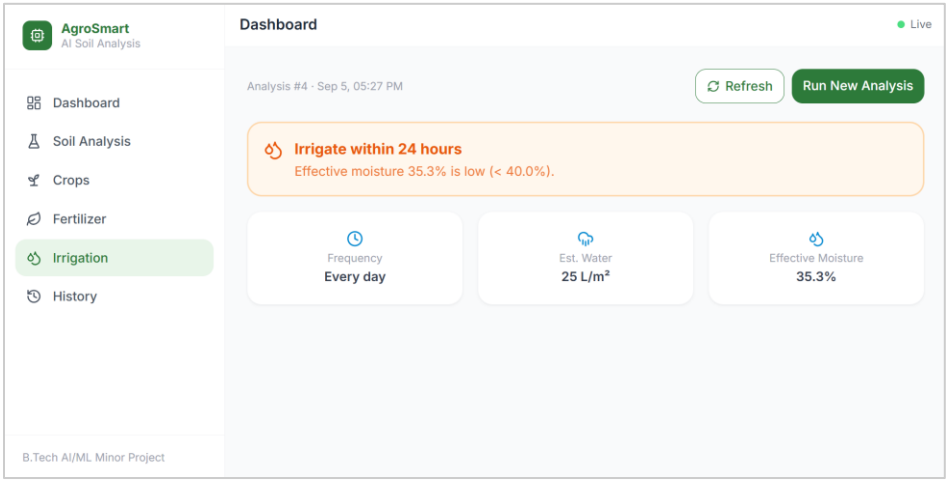


- **Live Sensor Dashboard**
- Shows the 8 live readings used by the models: moisture, pH, soil/air temperature, air humidity, N, P, K.

Work Done So Far — Results & History



- **Analysis History**
- Every run is logged: soil type, health score, top recommended crop, fertilizer, irrigation advice.



- **Irrigation Recommendation**
- Rule engine flags low effective moisture and suggests frequency and estimated water (L/m^2).

- **Note:** readings above are from the simulator (SENSOR_MODE=simulator); Fertilizer currently shows N/A as that model is still a prototype.

Work To Be Done

- Integrate and calibrate the real Raspberry Pi 5 sensor hardware (currently validated in simulator mode only).
- Run the full backend test suite and the evaluate_models.py script to report actual accuracy for the crop and soil-CNN models.
- Move the fertilizer model beyond its current Decision Tree prototype using the full Kaggle fertilizer dataset.
- Field-test the irrigation rule engine against real weather and soil conditions.
- Polish the React dashboard UI and prepare the system for the Part-II hardware demonstration.

Conclusion

Summary of Progress

- A working full-stack pipeline is in place: sensor/simulator input → Flask REST API → ML/rule-based analysis → React dashboard.
- All four decision components (crop, fertilizer, soil type, irrigation) are implemented end-to-end and produce results visible in the dashboard and history log.

Expected Outcome

- A low-cost, Raspberry Pi 5-based decision-support tool that a small farmer can use to get real-time, data-backed crop, fertilizer and irrigation guidance, to be hardware-validated and evaluated in Part-II.

References

[1] Z. Doshi, S. Nadkarni, R. Agrawal, and N. Shah, “AgroConsultant: Intelligent Crop Recommendation System Using Machine Learning Algorithms,” in Proc. IEEE ICCUBEA, Pune, India, 2018.

[2] K. Chatterjee, M. S. Obaidat, D. Samanta, B. Sadoun, S. H. Islam, and R. Chatterjee, “Classification of Soil Images Using Convolution Neural Networks,” in Proc. 2021 Int. Conf. Communications, Computing, Cybersecurity and Informatics (CCCI), 2021, pp. 1–5.

[3] E. K. Gyasi and S. Purushotham, “Soil-MobiNet: A Convolutional Neural Network Model Based Soil Classification to Determine Soil Morphology and Its Geospatial Location,” Sensors, vol. 23, no. 15, p. 6709, 2023.

[4] A. Ingle, “Crop Recommendation Dataset,” Kaggle, 2020. [Online]. Available: https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset

Thank You